Bayesian Optimal Stopping

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Bayesian Learning in Online Decision Making

Online decision-making under uncertainty raises a fundamental question about the relationship between *learning* and *decision-making*: when an algorithm begins interacting with an environment, should it

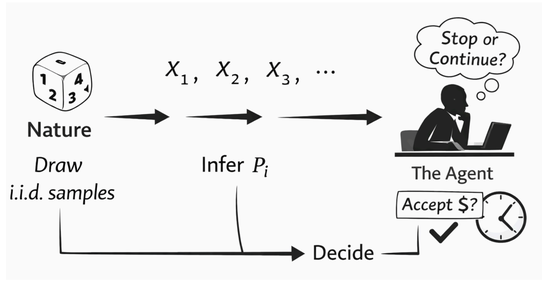
- rely on **prior knowledge** (training data),
- rely on **online samples**,
- or combine both in a **Bayesian** way?

While *be Bayesian* is often good advice, the value of prior knowledge versus online learning could vary dramatically, and in some settings, one of these sources of information is redundant for making the optimal decision.

Classical Models for *Beyond Worst-Case* Algorithms

Classical models for online algorithms under uncertainty sit at two opposite extremes and **often obscure the value of online learning**. This tension is especially sharp in **optimal stopping:** *the decision-maker observes a sequence of n i.i.d. samples X_t drawn from a prior distribution online and wants to stop on the largest possible observation*!

1. Assuming a known distribution **(prophet inequality)** is overly optimistic and, under this model, there is nothing to learn.
2. Adversarial models tend to be too pessimistic: For example, [1, Theorem 2]. showed that, for the optimal stopping problem under an unknown i.i.d. distribution, learning from early samples alone cannot improve on the benchmark of $1/e$ for when samples are exchangeable but completely adversarial otherwise **(secretary problem)**, where accepting the maximum so far after $t > n/e$ is optimal.



Bayesian Model

We introduce a Bayesian perspective to this landscape that interpolates smoothly between the classical known distributional model and models with no known or many competing possible distributions!

There are k known possible distributions to the decision-maker

$$\mathcal{P}_1, \dots, \mathcal{P}_k, \qquad \text{and a known, correct prior } \pi \in \Delta_k$$

over these candidates. Nature draws a distribution $I \sim \pi$, and conditioned on I, the decison-makers observes

$$X_t \overset{\text{i.i.d.}}{\sim} \mathcal{P}_I, \qquad t = 1, \dots, n.$$

single prediction — known i.i.d. prophet inequalities consistency CR: 0.745 — $k = 3$ — infinite/no predictions — unknown i.i.d. secretary robustness CR: $1/e$

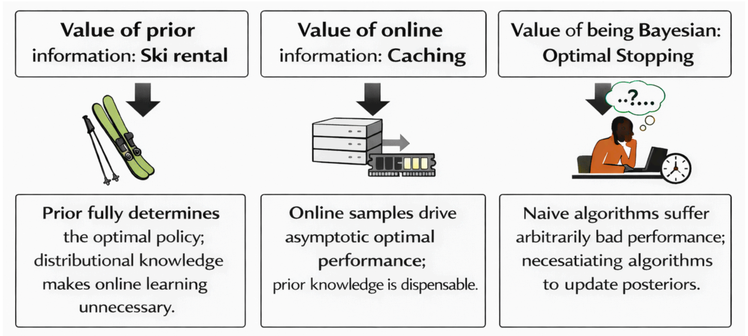
Performance Metrics: Competitive and Approximation Ratios

For an online algorithm ALG, let $\mathbb{E}_\pi[\text{ALG}]$ denote its expected value under prior π. We define

$$\text{CR}_\pi(\text{ALG}) := \frac{\mathbb{E}_\pi[\text{ALG}]}{\mathbb{E}_\pi[\text{OFF}]}, \qquad \text{AR}_\pi(\text{ALG}) := \frac{\mathbb{E}_\pi[\text{ALG}]}{\text{OPT}_\pi},$$

where OFF is the value of the best decision in hindsight, and OPT_π is the optimal expected value attainable by an online algorithm knowing π.

Different Regimes



- **Ski rental (prior only):** Let $T \in [n]$ be the random horizon with an arbitrary prior distribution and $\mathcal{H}_t$ be the information available at time t. Then, for every set $A \subseteq [n]$,

$$\Pr(T \in A \mid \mathcal{H}_t,\, T \geq t) = \Pr(T \in A \mid T \geq t).$$

- **Caching (samples only):** Under i.i.d. requests, a simple algorithm (using heavy-hitters methods) that behaves like the well-known LFU (Least Frequently Used) policy based only on online samples can drive asymptotically optimal performance.

Bayes-Optimal Dynamic Program → Approximation Algorithms

Define the *value-to-go* from state $(t, \mathcal{D}(t))$ recursively as

$$M_t(\mathcal{D}(t)) \;=\; \mathbb{E}_{X_{t+1} \sim \mathcal{P}_{\mathcal{D}(t)}}[\max\{X_{t+1},\; M_{t+1}(\mathcal{D}(t+1) \mid X_{t+1})\}]\,,$$

with $M_{n+1}(\mathcal{D}) = 0$. Here, $\mathcal{D}(t+1) \mid X_{t+1}$ denotes the posterior. The optimal online policy is therefore: at time t, given posterior $\mathcal{D}_{\text{belief}}(t)$, stop if and only if

$$X_t \;\geq\; M_t(\mathcal{D}_{\text{belief}}(t))\,.$$

The optimal policy jointly depends on the randomness of future observations and the induced stochastic evolution of posterior beliefs, which is **computationally intractable** to capture.

Research Question

This raises a **computational question**: must one compute the fully Bayesian optimal policy-often intractable (indeed, PSPACE-hard in general [3, Theorem 6])-or can *approximately Bayesian* algorithms achieve near-optimal performance efficiently?

Bayesian online is as *hard* as unknown online

By **Yao's minimax principle**,

$$\min_{\pi} \max_{\text{ALG}} \text{CR}_\pi(\text{ALG}) = \max_{\text{ALG}} \min_{\pi} \text{CR}_\pi(\text{ALG}).$$

This does not eliminate the possibility for an improvement in our Bayesian model when the support size k is finite.

Hardness → Need for Adaptivity

- **Theorem:** With k candidate i.i.d. distributions, a $1/e + 1/\text{poly}(k)$ competitive ratio is ***optimal and achievable***.
- This **partially resolves an open question** of Correa et al. [2] about improved guarantees for a finite set of candidate distributions. Our explicit hard instance also gives a **constructive proof** of the limiting $1/e$ upper bound in [2, Theorem 5.1].
- **Proposition:** No algorithm that ignores past observations can guarantee a positive constant approximation ratio.

Prior	Dist \ Val	1	n^3	n^6	n^9	$\cdots$	$n^{3(k-1)}$	n^{3k}
$1 - \sum_{i=1}^{k-1} \frac{1}{n^{3i}}$	$\mathcal{P}_1$	$1 - \frac{1}{n^2}$	$\frac{1}{n^2}$					
$\frac{1}{n^3}$	$\mathcal{P}_2$	$\left(1 - \frac{1}{n^2}\right)^2$	$\left(1 - \frac{1}{n^2}\right)\frac{1}{n^2}$	$\frac{1}{n^2}$				
$\frac{1}{n^6}$	$\mathcal{P}_3$	$\left(1 - \frac{1}{n^2}\right)^3$	$\left(1 - \frac{1}{n^2}\right)^2\frac{1}{n^2}$	$\left(1 - \frac{1}{n^2}\right)\frac{1}{n^2}$	$\frac{1}{n^2}$			
$\vdots$	$\vdots$							
$\frac{1}{n^{3(k-2)}}$	$\mathcal{P}_{k-1}$	$\left(1 - \frac{1}{n^2}\right)^{k-1}$	$\left(1 - \frac{1}{n^2}\right)^{k-2}\frac{1}{n^2}$	$\left(1 - \frac{1}{n^2}\right)^{k-3}\frac{1}{n^2}$	$\cdots$	$\left(1 - \frac{1}{n^2}\right)\frac{1}{n^2}$	$\frac{1}{n^2}$	
$\frac{1}{n^{3(k-1)}}$	$\mathcal{P}_k$	$\left(1 - \frac{1}{n^2}\right)^{k}$	$\left(1 - \frac{1}{n^2}\right)^{k-1}\frac{1}{n^2}$	$\left(1 - \frac{1}{n^2}\right)^{k-2}\frac{1}{n^2}$	$\cdots$	$\cdots$	$\left(1 - \frac{1}{n^2}\right)\frac{1}{n^2}$	$\frac{1}{n^2}$

Table: Probability Mass Functions for Distributions

Dual Martingales & Primal Algorithm

Let $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$ denote the observed history. A **dual martingale** is an integrable, adapted process $(Z_t)_t$ with

$$\mathbb{E}_\pi[Z_{t+1} \mid \mathcal{F}_t] = Z_t, \qquad Z_t \geq X_t \quad \text{a.s.}$$

Its terminal value $M = Z_n$ is a **dual certificate**, with $Z_t = \mathbb{E}_\pi[M \mid \mathcal{F}_t]$. For every stopping time $\tau \leq n$, *Doob's Optional Sampling Theorem* gives

$$\mathbb{E}_\pi[X_\tau] \leq \mathbb{E}_\pi[Z_\tau] = \mathbb{E}_\pi[M].$$

Thus $\mathbb{E}_\pi[M] \geq \text{OPT}_\pi$: the dual certifies an **upper bound on the optimal online algorithm**!

Primal improvement (Alg 1) Given a dual martingale $(Z_t)_t$ with certificate M and a baseline policy ALG satisfying

$$\mathbb{E}_\pi[\text{ALG}] \geq \alpha \mathbb{E}_\pi[M]$$

for $0 < \alpha < 1$, set $\gamma = g(\alpha) := \frac{1}{1+\ln(1/\alpha)}$.

1. With probability γ, run ALG.
2. Otherwise, draw $Q \in [\alpha, 1]$ with density $f_Q(q) = 1/(q \ln(1/\alpha))$ and accept the first $X_t \geq Q Z_t$.

The resulting policy ALG^+ satisfies

$$\mathbb{E}_\pi[\text{ALG}^+] \geq g(\alpha)\, \text{OPT}_\pi, \qquad g(\alpha) > \alpha.$$

Starting with the secretary policy gives a $1/2$**-approximation of the optimal online policy**!

Primal-Dual Algorithm

Iterative primal-dual improvement. Initialize ALG_0 as the secretary policy, $\alpha_0 = 1/e$, and $M_0 = \max_{t \leq n} X_t$. For $j = 0, \dots, J-1$, let $Z_t^{(j)} = \mathbb{E}_\pi[M_j \mid \mathcal{F}_t]$ and perform:

1. **Primal update:** Apply Alg 1 to $(\text{ALG}_j, Z^{(j)}, \alpha_j)$ to obtain ALG_{j+1}, and set $\alpha_{j+1} = g(\alpha_j)$.
2. **Dual update:** On each complete sample path, define

$$\rho_j = \max_{t \leq n} \frac{X_t}{Z_t^{(j)}}, \qquad M_{j+1} = M_j \max\{\alpha_j, \rho_j\}.$$

The dual remains feasible and shrinks: $M_{j+1} \leq M_j$. Every round preserves

$$\mathbb{E}_\pi[\text{ALG}_j] \geq \alpha_j \mathbb{E}_\pi[M_j] \geq \alpha_j \text{OPT}_\pi.$$

(a) Read one sample path **(b) Update that sample path**

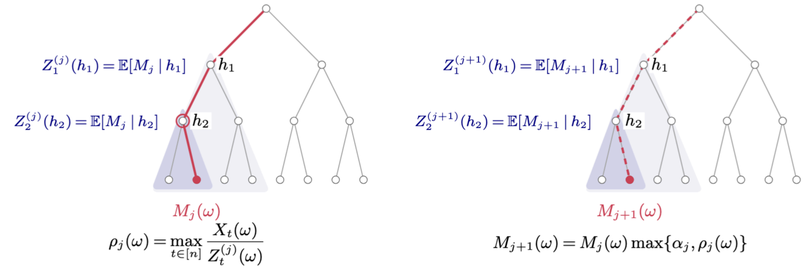


Figure: The pathwise dual update on the scenario tree

Polynomial-Time Approximation Scheme

Return ALG_J after $J = O(1/\varepsilon)$ rounds. Using suitably accurate importance-weighted Monte Carlo estimates, $\text{AR}_\pi(\text{ALG}_J) \geq 1 - \varepsilon$. For each fixed $\varepsilon > 0$, the implementation runs in polynomial time given the finitely supported candidate PMFs.

References

[1] José Correa, Paul Dütting, Felix Fischer, and Kevin Schewior, *Prophet inequalities for i.i.d. random variables from an unknown distribution*, Proceedings of the 2019 ACM Conference on Economics and Computation (New York, NY, USA), EC '19, Association for Computing Machinery, 2019, p. 3–17.

[2] José Correa, Paul Dütting, Felix Fischer, Kevin Schewior, and Bruno Ziliotto, *Unknown i.i.d. prophets: Better bounds, streaming algorithms, and a new impossibility*, 2020.

[3] Christos H. Papadimitriou and John N. Tsitsiklis, *The complexity of markov decision processes*, Mathematics of Operations Research (1987), no. 3, 441–450.